



Kvaerner Oil & Gas

Project : **Huldra EPC Wellhead Platform Topsides**

Title : **Stystem Control Diagram, Air Compressor A Sequences**

Doc. No : **C025-W-H063-J-*E-005-01**

KOG Project No : **C131**

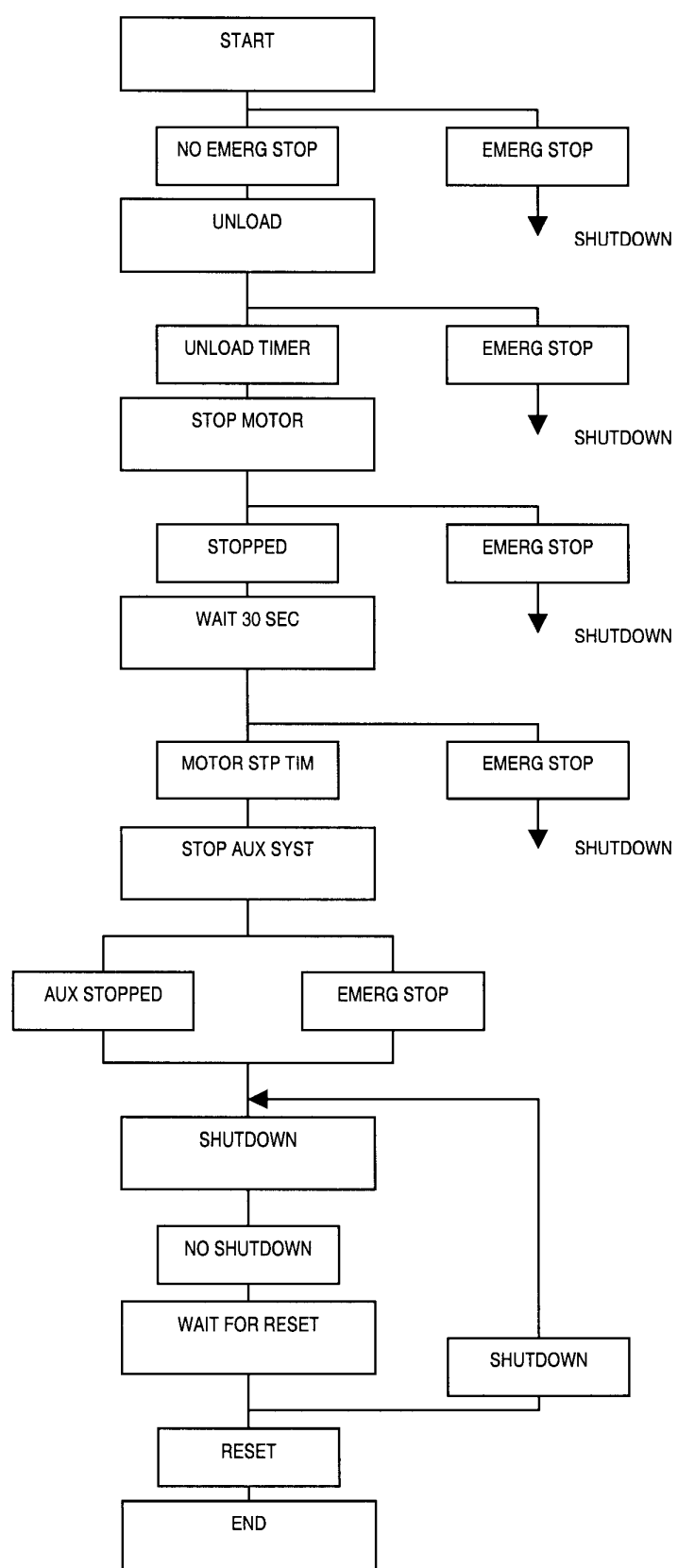
Statoil Project No : **C025**

A2	24.07.03	As Built AO600126	NH-J	SMB/RT	DB
A1	30.09.01	AS BUILT	EiL	<i>AB.</i>	<i>JP</i>
0	20.03.01	ISSUED FOR CONSTRUCTION	Buk		
01D	30.11.00	ISSUED FOR DESIGN	Buk		
Revision	Date	Revision Description	Prepared	Checked	Approved

INPUTS		
Document Number	Title	Status
C025-W-H063-J-SX-001 Comp Air: E7100-C-02	Functional description air distribution system 63 Air Cooled DRYCLON CONTROLLER, Chemitrol/K.O.G./STATOIL (Huldra)	04D A2

REVISIONS & HOLDS	
Revision	Description of Changes & Holds
01D	Issued for design
0	Issued for construction – Statoil comments incorporated
A1	As Built
A2	As Built AO600126





COMPRESSOR A SEQUENCE

Phase	Step	Actions	Outputs to	Next step	Wait for (before entering next step)	Jump condition.	Jump to step	Remarks
START	0	N.A.	N.A.	NEXT	N.A	N.A	N.A	**
Start inhibit	1	All equipment de-energized	Stop signal to: 63-KC 50A 63-KE 50A 63-KE 50B 63-PE 50 Close signal to: 63-XV 2022 63-HV 2037 Set 63-KE 50B to manual	NEXT	Available and BCL from equipment: 1: 63-KC 50A.XE AND 2: 63-KE 50A.XE AND 3: 63-KE 50B.XE AND 4: 63-PE 50.XE AND 5: 63-KC 50A.BCL AND 6: 63-KE 50A.BCL AND 7: 63-PE 50.BCL AND = TRUE	N.A.	N.A.	*
Ready for start	2	All equipment de-energized	Stop signal to: 63-KC 50A 63-KE 50A 63-KE 50B 63-PE 50 Close signal to: 63-XV 2022 63-HV 2037 Set 63-KE 50B to manual	NEXT OR JUMP 1 OR JUMP 2	Start command from OS or logic: 63-HS 9000.Y0 OR (63-PI 2000.BXL AND Z258.Y) OR (63-HS 9001L.Y AND Z259.Y) =TRUE	JUMP 1:Equipment not available: 1: 63-KC 50A.XE AND 2: 63-KE 50A.XE AND 3: 63-KE 50B.XE AND 4: 63-PE 50.XE AND 5: 63-KC 50A.BCL AND 6: 63-KE 50A.BCL AND 7: 63-PE 50.BCL AND = FALSE JUMP2: Stop command from OS 63-HS 9000.Y1 = TRUE	JUMP1: 1 (Start inhibit) JUMP2: 14 (Go to SD sequence)	*
Standby	3	Aux. equipment de-energized	Stop signal to: 63-KE 50A 63-KE 50B 63-PE 50 Close signal to: 63-XV 2022 63-HV 2037 Set 63-KE 50B to manual	NEXT OR JUMP	Low pressure: 63-PI 2000 BXL=TRUE	Stop command from OS: 63-HS 9000.Y1=TRUE	2 (Ready for start)	*
Start aux.syst	4	Start oil pump, open discharge valve, start fan 1 and set fan 2 to auto.	Start signal to: 63-KE 50A AND 63-KE50B 63-PE 50 Open signal to: 63-HV 2037 Set 63-KE 50B to auto	NEXT OR JUMP	Aux. Syst confirmed started: 63-PE 50.BCH AND 63-KE 50A OR 63-KE50B AND 63-PI 2025.BXH = TRUE	Stop command from OS: 63-HS 9000.Y1=TRUE	2 (Ready for start)	*
Start motor	5	Inhibit vibration alarms Start Compressor A main drive	63-YAH 2038A/B.FU and 63-YAHH 2038A/B.FU = TRUE (via Z305) Start sinal to: 63-KC 50A	NEXT OR JUMP	Motor running: 63-KC 50A.BCH = TRUE	Stop command from OS: 63-HS 9000.Y1=TRUE	12 (Stopping not available)	*
Load delay	6	Start Load Delay Timer– 2 sec	Start timer T1157 (Z157.X)	NEXT OR JUMP	Load delay expired: T1157(Z157.Y) = TRUE	Stop command from OS: 63-HS 9000.Y1=TRUE	12 (Stopping not available)	
Offload	7	De-energize load valve Turn off inhibit of vibration alarm Start Run On Timer – 15 m	Close signal to: 63-XV 2022 63-YAH 2038A/B.FU and 63-YAHH 2038A/B.FU = FALSE (via Z305) Start timer T1158 (Z158.X)	NEXT OR JUMP 1 OR JUMP 2	Low pressure and discharge valve confirmed open and, load timer or reload timer: 63-PI 2000 .BXL AND 63-HV 2037.BCH AND (T1157(Z157.Y) OR T1160(Z160.Y)) = TRUE	JUMP 1: Run On Timer: T1158(Z158.Y) = TRUE JUMP 2: Stop command from OS: 63-HS 9000.Y1=TRUE	JUMP 1: 9 (Stopping available) JUMP 2: 11 (Unloading)	*

Onload	8	Energize load valve Start reload timer on exit from step.	Open signal to: 63-XV 2022 Start timer T1160 (Z160.X) on exit from step	JUMP 1 OR JUMP 2	N.A.		JUMP 1: High press or Discharge Valve NOT confirmed open: 63-PI 2034.BXH = TRUE OR 63-HV 2037.BCH = FALSE	JUMP 1: 7 (Offload) JUMP 2: 11 (Unloading)	*
Stopping available	9	Inhibit vibration alarms Stop Compressop A main drive	63-YAH 2038A/B.FU and 63-YAHH 2038A/B.FU = TRUE (via Z305) Stop signal to: 63-KC 50A	NEXT	Motor stopped: 63-KC 50A.BCL = TRUE		JUMP 2: Stop command from OS: 63-HS 9000.Y1=TRUE N.A.		*
Motor Stop Timer a – Wait 30 sec	10	Start Motor Stop Timer – 30 sec Turn off inhibit of vibration alarm on exit from step	Start timer T1159 (Z159.X via Z329) 63-YAH 2038A/B.FU and 63-YAHH 2038A/B.FU = FALSE (via Z305)	JUMP 1 OR JUMP 2	N.A.		JUMP 1: Motor stop timer expired: T1159(Z159.Y) = TRUE AND 63-HS 9000.Y1 = FALSE JUMP 2: Stop command from OS: 63-HS 9000.Y1=TRUE	JUMP 1: 3 (Standby) JUMP 2: 12 (Stopping not available)	
Unload	11	De-energize load valve and start Start Unload Timer – 8 sec.	Close signal to: 63-XV 2022 Start timer T1161 (Z161.X)	NEXT OR JUMP	Unloading time expired: T1161(Z161.Y) = TRUE		Start command from OS: 63-HS 9000.Y0=TRUE	7 (Offload)	*
Stopping not available	12	Inhibit vibration alarms Stop Compressop A main drive	63-YAH 2038A/B.FU and 63-YAHH 2038A/B.FU = TRUE (via Z305) Stop signal to: 63-KC 50A	NEXT	Motor stopped: 63-KC 50A.BCL = TRUE	N.A.		N.A.	*
Motor Stop Timer b	13	Start Motor Stop Timer – 30 sec Turn off inhibit of vibration alarm on exit from step	Start timer T1159 (Z159.X via Z329) 63-YAH 2038A/B.FU and 63-YAHH 2038A/B.FU = FALSE (via Z305)	JUMP 1 OR JUMP 2	N.A.		JUMP 1: Start command from OS: 63-HS 9000.Y0=TRUE JUMP 2: Motor stop time expired: T1159(Z159.Y) = TRUE	JUMP 1: 9 (Stopping available) JUMP 2: 1 (Start inhibit)	
Go to SD sequence	14	Start 63-SD_COMP_A	Start command to 63-HS 9000L	END	N.A.	N.A.		N.A.	
End	15								

* When a start/ stop or open/close command is given to an object, an auto command is also given.

** The sequence is started automatically upon start of the PLC. It will run as long as the shutdown sequence is not running. I.e. the compressor sequence is aborted when the shutdown sequence is started. When the shutdown sequence is terminated the compressor sequence is restarted.

Chart Properties\ Operating Parameters:

Defaults:

Step Control:	T
Operating Mode:	Auto
Command output:	Yes
Cyclic operation:	Yes
Time monitoring:	No
Start Chart:	
Autostart:	Yes
Use default settings when SFC start:	Yes

SHUTDOWN SEQUENCE COMPRESSOR A

Phase	Step	Actions	Outputs to	Next step	Wait for (before entering next step)	Jump condition	Jump to step	Remarks
START	0	N.A.	N.A.	NEXT OR JUMP	NOT Emergency stop: Z231.OUT = FALSE	Emergency Stop Activated: Z231.OUT = TRUE	5 (Shutdown)	The sequence is started by Z172.OUT See SCD C025-W-H063-J-*E-002-01
Unload	1	Stop Comp_A sequence De-energize load valve Start Unload Timer – 2 sec	Stop command to 63-HS 9000H Close signal to 63-XV 2022 Start timer T1162 (Z162.X)	NEXT OR JUMP	Unload timer - shutdown : Z162.Y	Emergency Stop Activated: Z231.OUT = TRUE	5 (Shutdown)	*
Stop motor	2	Inhibit vibration alarms Stop Compressop A main drive	63-YAH 2038A/B.FU and 63-YAHH 2038A/B.FU = TRUE (via Z305) Stop signal to 63-KC 50A	NEXT OR JUMP	Main drive motor confirmed stopped: 63-KC 50A.BCL = TRUE	Emergency Stop Activated: Z231.OUT = TRUE	5 (Shutdown)	*
Motor Stop Timer- Wait 30 sec.	3	Start Motor Stop Timer – 30 sec Turn off inhibit of vibration alarm on exit from step	Start timer T1159 (Z159.X) (via Z329) 63-YAH 2038A/B.FU and 63-YAHH 2038A/B.FU = FALSE (via Z305)	NEXT OR JUMP	Motor Stop Timer : Z159.Y	Emergency Stop Activated: Z231.OUT = TRUE	5 (Shutdown)	
Stop Aux. syst	4	Stop oil pump, close discharge valve, stop fans and set fan 2 to manual	Stop signal to: 63-PE 50 63-KE 50A 63-KE 50B Close signal to: 63-HV 2037 Set 63-KE 50B to manual	NEXT	Aux. system confirmed stopped or emergency stop: 63-KE 50A.BCL and 63-KE 50B.BCL and 63-PE 50.BCL and 63-HV 2037.BCL = TRUE OR Z231.OUT = TRUE	N.A.	N.A.	*
Shutdown	5	Stop Comp_A sequence	Stop command to 63-HS 9000H	NEXT	No emergency stop and no Shutdown command: Z172.OUT AND Z231.OUT = FALSE	N.A.	N.A.	*
Wait for reset	6	Reset command from OS	N.A.	NEXT OR JUMP	63-HS 9000.Y2 = TRUE	Emergency stop or Controlled shutdown command: Z231.OUT OR Z172.OUT = TRUE	5 (Shutdown)	
End	7	Start Comp_A sequence	Start command to 63-HS 9000H	END	N.A.	N.A.	N.A.	*

* When a start/ stop or open/close command is given to an object, an auto command is also given.

Chart Properties\ Operating Parameters:

Defaults:

Step Control:	T
Operating Mode:	Auto
Command output:	Yes
Cyclic operation:	No
Time monitoring:	No
Start Chart:	
Autostart:	Yes
Use default settings when SFC start:	Yes